LAB 06

# Strategy Prototype: TradingView Strategy Tester, Paper Trading from the Chart and Pipeline Reconciliation

| Item | Detail |
|---|---|
| Topic | Topic 2: Technical Indicators, Trading Strategies and Metrics with TradingView and Alpaca |
| Pipeline stage | Strategy prototype |
| TSC reference | A2, A3, K3, K7 |
| Duration | 30 min |
| Folder | labs/Lab06-pine-strategy-tester/ |
| Files | Lab06-pine-strategy-tester/reconcile_stage.py, Lab06-pine-strategy-tester/lcc_ma_crossover_strategy.pine, Lab06-pine-strategy-tester/tv_strategy_results.yaml, pipeline/reconcile.py, Lab06-pine-strategy-tester/solution/reconcile.py |
| Mode | Runs ONLINE with your Alpaca paper keys in labs/.env, or OFFLINE on the bundled data and the paper-broker simulator. Paper trading only. |

*Lion City Capital and every scenario in these materials are fictitious and for training only. All trading in this course is paper (simulated) trading. Nothing in these materials is investment advice or a recommendation to trade any security.*

## Scenario

A portfolio manager prototypes a golden-cross rule for the desk's SPY sleeve in TradingView. Before it can enter the automated pipeline, the Strategy Tester result (with commission and slippage) must be reproduced by an independent pipeline backtest, and the trader must be able to put the idea into action on a paper account straight from the chart.

## Objective

Configure a Pine Script v6 strategy with costs, read the Strategy Tester, build `python -m pipeline reconcile` to check the Pine settings and reproduce the results, and place and manage paper trades from the chart with TradingView Paper Trading (and optionally an Alpaca paper account).

## Before You Start

- Open a terminal in the labs folder and activate your virtual environment.
- Have your AI coding assistant open (for example Claude Code) in the labs folder if you want to rebuild the code from the prompts; the finished reference code is already in this folder.
- Record every result you are asked to check - the same evidence is asked for in the Practical Performance assessment.

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## Step-by-Step Instructions

### Step 1: Read the strategy code (A2, K3)

Open Lab06-pine-strategy-tester/lcc_ma_crossover_strategy.pine. Identify the strategy() settings (initial_capital 100000, 100% of equity, commission_type cash_per_order 1.0, slippage 2, pyramiding 0, process_orders_on_close false) and the rules (ta.crossover / ta.crossunder of SMA 50 and SMA 200).

| AI prompt - paste into your AI coding assistant |
| --- |
| Explain labs/Lab06-pine-strategy-tester/lcc_ma_crossover_strategy.pine for a finance professional: each strategy() setting, when an order is filled (process_orders_on_close = false means the NEXT bar's open), and how commission and slippage reduce the result. Do not change the file. |

*Check: You can state that a signal at Monday's close is filled at Tuesday's open, with USD 1 commission per order.*

### Step 2: Run it in the Strategy Tester and record the results (A3, K7)

Open AMEX:SPY on 1D, paste the script into a new Pine Editor tab, save and Add to chart. Open the Strategy Tester panel and copy net profit, closed trades, percent profitable, profit factor, max drawdown, Sharpe, Sortino and commission paid into Lab06-pine-strategy-tester/tv_strategy_results.yaml (replace the nulls).

*Check: The chart shows SMA 50 / SMA 200, green shading while in a position, and the Performance Summary tab.*

### Step 3: Build the reconciliation stage (A2, A3, K3)

Ask your assistant for pipeline/reconcile.py.

| AI prompt - paste into your AI coding assistant |
| --- |
| Create labs/pipeline/reconcile.py. parse_pine(path) reads the //@version, the keyword settings inside strategy(...) and the input.int defaults (fastLen, slowLen) from Lab06-pine-strategy-tester/lcc_ma_crossover_strategy.pine. settings_checks(pine, cfg) returns PASS/FAIL rows: version 6; process_orders_on_close = false (next-bar-open fills = config strategy.execution next_open); commission_type cash_per_order with the value in config costs.tradingview; slippage > 0; default_qty_value 100 (% of equity); pyramiding 0. run(symbol='SPY', fast=None, slow=None) backtests the same rule with common.backtest.backtest(bars, cross_signal(ta_sma(fast), ta_sma(slow)), costs=pipeline.config.tradingview_cost_model(cfg), execution='next_open') and buy_and_hold, prints the Strategy Tester equivalents (net profit USD and %, closed and open trades, percent profitable, profit factor, max drawdown, Sharpe, Sortino, commission, buy and hold) next to the values in Lab06-pine-strategy-tester/tv_strategy_results.yaml with a difference column, the trade list, and four likely causes of gaps (adjusted vs unadjusted prices, 2018 history start, whole vs fractional shares, ticks vs bps slippage). Write _outputs/Lab06_reconciliation.yaml (plain Python numbers) and chart Lab06_crosscheck.png with common.plotting.plot_equity_and_drawdown. Exit code 1 if a settings check fails. Rules: run from the labs folder; reuse labs/common (config, data, broker, indicators, metrics, backtest, plotting) and earlier pipeline stages instead of rewriting them; do not modify common/ or data/; read every setting from pipeline/config.yaml via pipeline.config.load_config(); never put API keys or secrets in code; paper trading only; keep a main(argv) function so `python -m pipeline <command>` works; add short comments a finance professional can follow. |

*Check: python -m pipeline reconcile --help shows --symbol, --fast and --slow.*

### Step 4: Reconcile TradingView with the pipeline (A2, A3)

Run: python -m pipeline reconcile --symbol SPY --fast 50 --slow 200 (same as: python3 Lab06-pine-strategy-tester/reconcile_stage.py).

*Check: 6 settings checks PASS. Pipeline: net profit USD 79,878 (+79.88%), 4 closed trades + 1 open, 50.0% profitable, profit factor 5.03, max drawdown -33.72%, Sharpe 0.55, Sortino 0.76, commission USD 9.00, buy and hold +187.26%; first entry 2018-10-17; your TradingView figures appear in the TradingView column with differences.*

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### Step 5: Explain the differences (A3)

Compare the two trade lists. Use the causes printed by the stage to explain each gap in one sentence.

| AI prompt - paste into your AI coding assistant |
| --- |
| Here is my reconciliation output: <paste the table from python -m pipeline reconcile>. For each metric where TradingView and the pipeline differ by more than 5%, say which of these causes is most likely and how I could confirm it: adjusted vs unadjusted prices, the 2018 data start and SMA 200 warm-up, whole vs fractional shares, slippage in ticks vs bps. |

*Check: Each gap is linked to a named cause; typically TradingView's longer history changes the first trade.*

### Step 6: Stress the costs and the speed of the rule (K7)

In TradingView change commission to 0.1% of order value, slippage to 5 ticks and the inputs to 20/50. Then run: python -m pipeline reconcile --fast 20 --slow 50.

*Check: Pipeline SMA 20/50: net profit USD 88,228 (+88.23%), 16 closed trades, max drawdown -27.84%, Sharpe 0.72, commission USD 33 - more trades pay more cost; record how TradingView's net profit falls with 0.1% commission.*

### Step 7: Put the strategy into action with TradingView Paper Trading (A3)

Open the Trading Panel at the bottom of the chart and choose 'Paper Trading by TradingView', then Connect. In the account settings set the starting balance to 100,000 USD and a commission (for example USD 1 per order). From the chart (Buy button, or right-click > Trade) place: (1) a market buy of 10 SPY, (2) a limit buy 2% below the last price, (3) a buy with take-profit and stop-loss (bracket) in the order ticket.

| AI prompt - paste into your AI coding assistant |
| --- |
| Act as my execution checklist for TradingView Paper Trading on AMEX:SPY with USD 100,000. Using the last pipeline close of 676.64, give the order-ticket fields and prices for (1) a market buy of 10 shares, (2) a limit buy 2% below the last price, (3) a bracket buy with take-profit +5% and stop-loss -3%; then say which status each order should show in the Orders and Positions tabs. |

*Check: Market buy filled in Positions; limit buy (about 663.11) Working in Orders; bracket with take-profit about 710.47 and stop-loss about 656.34 shown as lines on the chart; commission deducted in the account summary.*

### Step 8: Manage the paper position from the chart (A3, K7)

Drag the stop-loss line to move it, close half the position, then close all. Read the History tab and the account balance. Note: a Pine strategy does not send orders to the Trading Panel by itself - automated orders go through alerts and webhooks (Lab 17) or the pipeline's execution stage (Lab 8).

*Check: History lists every fill with its commission; the position is flat and the balance reflects P&L and costs.*

### Step 9: Optional - connect the Alpaca paper account in TradingView (A3)

In the Trading Panel choose Alpaca, log in, select the PAPER account and place one small market or limit order from the chart. Then run python -m pipeline check (online) to see the same position from the pipeline.

*Check: The order appears in the Alpaca paper dashboard and as an open position in section [4] of the check.*

### Step 10: Decide whether to promote the rule (A2)

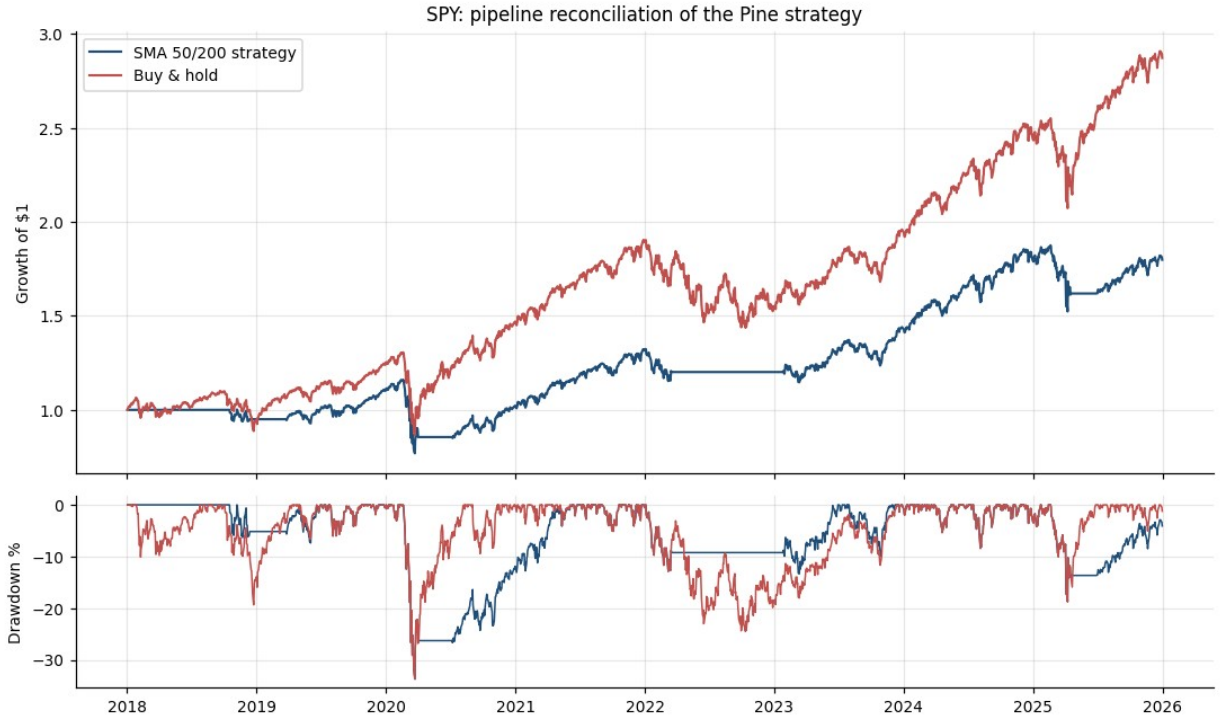
Apply the strategy to SGX:D05 and compare. Decide whether SMA 50/200 should replace the pipeline's configured ema_cross 10/30 (do not change config.yaml yet - Labs 9 and 10 measure it properly).

*Check: A one-paragraph decision that cites net profit vs buy and hold (+79.88% vs +187.26%) and the drawdown.*

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## Expected Output

python -m pipeline reconcile --symbol SPY --fast 50 --slow 200: 6 settings checks PASS; net profit USD 79,878 (+79.88%), 4 closed trades + 1 open, 50.0% profitable, profit factor 5.03, max drawdown -33.72%, Sharpe 0.55, Sortino 0.76, commission USD 9.00 versus buy and hold +187.26%; SMA 20/50: +88.23%, 16 trades, commission USD 33; TradingView Paper Trading shows a filled market order, a working limit order and a bracket; files _outputs/Lab06_reconciliation.yaml and chart _outputs/Lab06_crosscheck.png.




*Lab 6 output (Lab06_crosscheck.png), offline mode on the bundled data.*

## Guide: Paper Trading on TradingView, Step by Step

TradingView's built-in Paper Trading account lets you place simulated orders from the chart. You use it in Lab 6 to put the moving-average strategy into action.

### 1. Connect Paper Trading

On Supercharts click Trade (top right). Choose Paper Trading from the broker list and click Connect. The Trading Panel opens at the bottom of the chart.

### 2. Set your starting capital and commission

In the Trading Panel open the account settings and set the starting balance and a commission that matches the course cost assumptions, so paper results are comparable with the Python backtest.

### 3. Open the order ticket

Click Buy or Sell (or Trade) to open the order ticket. Choose the side, quantity and order type - Market, Limit or Stop - and the price for a limit or stop order.

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### 4. Add a stop loss and take profit

In the order ticket's risk settings tick Stop Loss and Take Profit and enter the levels (for example 2 x ATR below and above the entry). Then place the order.

### 5. Watch the order and position

Use the Trading Panel tabs: Positions (open trades and unrealised P&L), Orders (working, filled, cancelled) and Account History. Orders and positions also appear as lines on the chart, which you can drag to modify.

### 6. Put the strategy into action

With the Lab 6 Pine strategy on the chart, read its latest signal in the Strategy Tester's List of Trades and place the matching paper order. Record the fill price and compare it with the strategy's simulated fill.

### 7. Close and reset

Close a position from the Positions tab (or the x on its chart line). To start again, reset the paper account from the Trading Panel's account settings - this clears its orders and history.

*Paper fills use TradingView's prices but cannot show your own market impact - compare them with the backtest's assumed costs.*

## Debrief

Choosing a rule and its parameters is a modelling decision (A2); the Strategy Tester and TradingView Paper Trading are analytics and execution platforms (A3) that report standard metrics (K3). An independent reconciliation is the gate between a chart prototype and the automated pipeline, and costs typed into strategy() (K7) barely touch a slow rule but compound for a fast one.

## Troubleshooting

| Symptom | What to do |
|---|---|
| Strategy Tester shows 'No data' | Check the backtest dates in the inputs and that the chart is 1D. |
| Different first trade | Expected: the bundled data starts in 2018, so the pipeline's SMA 200 exists only from October 2018. |
| A settings check FAILs | Someone changed the Pine file or config.yaml costs.tradingview; make them agree and rerun. |
| Paper Trading not listed | Open the Trading Panel (bottom toolbar) and scroll the broker list; it is called 'Paper Trading by TradingView'. |

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